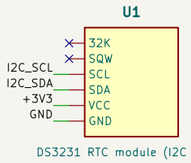
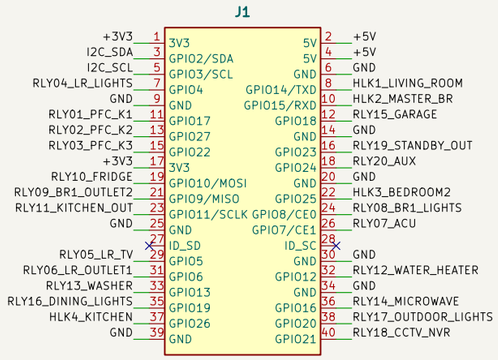
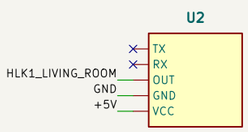
## RASPBERRY PI 5 DOMAIN  (relays, presence sensors, RTC)



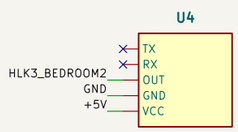

DS3231 RTC module (I2C 0x68)



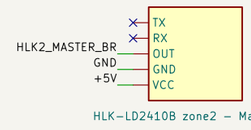

Raspberry Pi 5 — 40-pin GPIO header



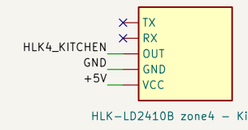

HLK-LD2410B zone1 — Living Room



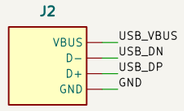

HLK-LD2410B zone3 — Bedroom2



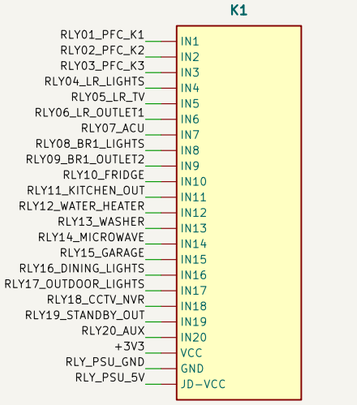

HLK-LD2410B zone2 — Master BR



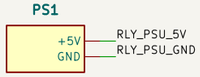

HLK-LD2410B zone4 — Kitchen




Raspberry Pi 5 — USB-A port




20-ch opto relay board (active-LOW)




5V >=2A relay coil supply

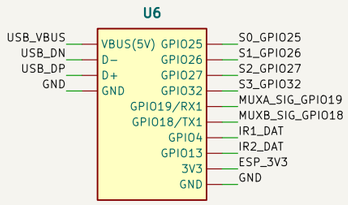
WIRING NOTES
1. Same label name = same wire. Connect every pin that shares a label.
   Full pin-to-pin list: NETLIST.md
2. J1 pin numbers are PHYSICAL header pins (not BCM).
3. K1 relay board: REMOVE the VCC/JD-VCC jumper.
   VCC = Pi 3V3 (opto LEDs); JD-VCC + GND = PS1 (5V >=2A).
   Do NOT tie Pi GND to K1 GND (keeps opto isolation).
4. Relays are active-LOW: GPIO LOW = relay ON.
   RLY01-03 drive the PFC capacitor-bank contactors.
5. GPIO14/15 (HLK zones 1-2) are UART pins:
   disable serial console + serial hardware.
6. Keep SPI (GPIO7-11) and audio/PWM overlays
   (GPIO12/13/18/19) disabled.
7. Boot check: GPIO9-27 pull down at power-up, so relays
   may click ON until app.py starts. Fix if needed:
   config.txt: gpio=9-13,16-24,27=op,dh
8. ESP32 side is all 3.3V: PZEM TTL '5V' pins and mux
   VCC come from ESP32 3V3 (ESP_3V3). No level shifter.
9. Two muxes, same S0-S3 wires: Mux A sends the selected
   PZEM TX to ESP32 RX1, Mux B sends ESP32 TX1 to that
   PZEM RX only. Channel Cn = PZEM #n on BOTH muxes.
10. Both mux EN = GND (always on). C11-C15 unused.
    IR modules: VCC = ESP32 5V/VIN (USB_VBUS).
11. Control wiring only: mains side not shown.

### MUX WIRING  (2x CD74HC4067, Mux A + Mux B)

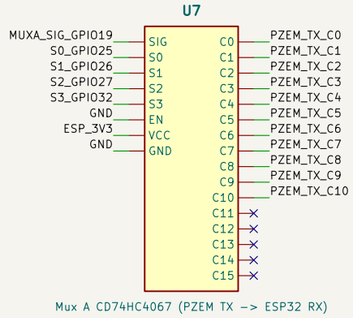
| Mux pin | Mux A (U7) | Mux B (U19) |
|---|---|---|
| S0 | ESP32 GPIO25 | ESP32 GPIO25  (same wire) |
| S1 | ESP32 GPIO26 | ESP32 GPIO26  (same wire) |
| S2 | ESP32 GPIO27 | ESP32 GPIO27  (same wire) |
| S3 | ESP32 GPIO32 | ESP32 GPIO32  (same wire) |
| SIG | ESP32 GPIO19 (RX1) | ESP32 GPIO18 (TX1) |
| VCC | ESP32 3V3 | ESP32 3V3 |
| GND | ESP32 GND | ESP32 GND |
| EN | GND | GND |
| C0-C10 | TX of PZEM #0-#10 | RX of PZEM #0-#10 |
| C11-C15 | open | open |

- S0-S3, VCC and GND go to BOTH muxes: one wire from the ESP32,
  then jump Mux A S0 -> Mux B S0 (or split it with a Y splice).
  Same for S1, S2, S3, VCC, GND/EN. Only SIG is different.
- Channel Cn = PZEM #n on BOTH muxes: its TX on Mux A Cn,
  its RX on Mux B Cn. Wrong channel = that PZEM shows STANDBY.
- Both muxes switch together: ESP32 asks PZEM #n through Mux B,
  PZEM #n answers through Mux A.

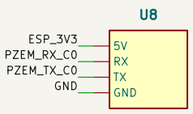
## ESP32 DOMAIN  (11x PZEM-004T via 2x CD74HC4067, 2x aircon IR, 3.3V logic)



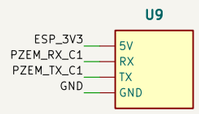

ESP32 DevKit (USB to Pi)



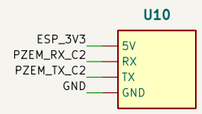

Mux A CD74HC4067 (PZEM TX -> ESP32 RX)



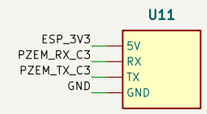

PZEM #0 Main Panel



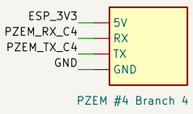

PZEM #1 Branch 1 (R4)



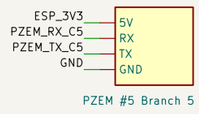

PZEM #2 Branch 2 (R5)



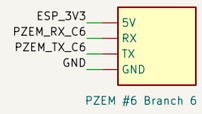

PZEM #3 Branch 3 (R6)



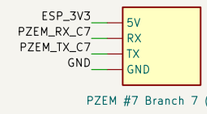

PZEM #4 Branch 4 (R7)



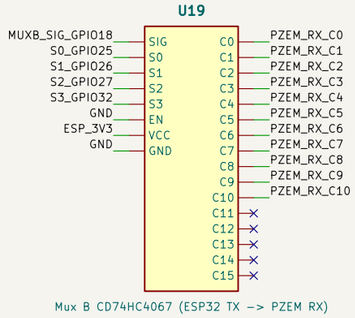

PZEM #5 Branch 5 (R8)



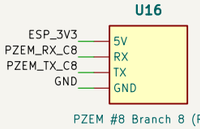

PZEM #6 Branch 6 (R9)



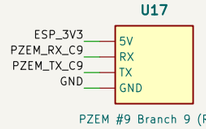

PZEM #7 Branch 7 (R10)



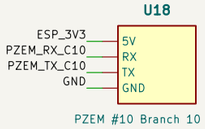

Mux B CD74HC4067 (ESP32 TX -> PZEM RX)



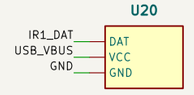

PZEM #8 Branch 8 (R11)



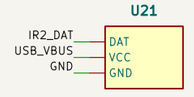

PZEM #9 Branch 9 (R12)




PZEM #10 Branch 10 (R13)




IR transmitter 1 — Room 1 aircon (AUX)




IR transmitter 2 — Room 2 aircon (Panasonic)

Control (low-voltage) wiring only
Generated by generate_schematic.py from app.py + ESP32 .ino
**REMS / RECMS Thesis**

Sheet: /
File: rems-control-wiring.kicad_sch

**Title: REMS — Control Wiring Schematic**

| Size: A3 | Date: 2026-10-30 | **Rev: 1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 1/1 |